Pre-questionnaire – Computational Notebook 2

(completed on LMS, digital format)

- In your own words, define a random variable

- Provide an example of a random variable

- Have you previously used/defined random variables in another course or contexts?
  - No
  - Yes, in statistics
  - Yes, in another engineering course
  - Yes, in a math course

  If yes, describe briefly (short response)

- Consider the following statement: "*The number of defective components in a batch of 100 units*"
  Do you think this is a random variable?
  - Yes
  - No
  - Not sure

  Explain your reasoning (short response)

- Rate your confidence in the following (1 lowest to 5 highest):

	1	2	3	4	5
I can define a random variable (RV)					
I can identify examples of random variables					
I can distinguish between discrete and continuous RVs					
I can connect problem statements to RVs					
I can use MATLAB for operations with RVs					
I can use RVs to learn about data trends					

- What do you expect to learn from this notebook?

Post-questionnaire – Computational Notebook 2

(completed on LMS, digital format)

- In your own words, what is a random variable?
- What key features distinguish/make a random variable?
- What is the difference between a discrete and a continuous random variable?
- Provide one example of a DRV
- Provide one example of a CRV
- Provide an example of when a continuous random variable can be used to approximate the behavior of a discrete random variable.
- What new insight about random variables did you learn from this notebook?

- Rate your confidence in the following (1 lowest to 5 highest):

	1	2	3	4	5
I can define a random variable (RV)					
I can identify examples of random variables					
I can distinguish between discrete and continuous RVs					
I can connect problem statements to RVs					
I can use MATLAB for operations with RVs					
I can use RVs to learn about data trends					

- What aspects of this notebook were most helpful for your understanding of RVs?
- What aspects of this notebook were least helpful or confusing?
- Is there any concept from the notebook that you still struggle with?
- Compared to traditional lectures, what aspects of notebook-based learning do you find most engaging?
- How much time did you spend working through the notebook?
  (radial buttons)
  - < 15 min
  - 15-30 min
  - 30-45 min
  - 45-60 min
  - > 60 min

- Notebook Evaluation (1 - strongly disagree to 5 - strongly agree):

	1	2	3	4	5
The notebook improved my understanding of RVs					
The notebook helped me understand differences between DRVs and CRVs					
Visualizations and examples helped me understand the concepts					
The examples were relevant and useful					
MATLAB activities reinforced the theory					
The notebook was easy to follow					